



EMERGING WEB TECHNOLOGIES

Gesture-Based Web Hand Tracking

Using your hands to control the web — no touch required

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What We'll Cover



01

Definition & Description

What is hand tracking? How does it work?



02

Examples & Applications

Real-world web use cases



03

Tools & Libraries

MediaPipe, TensorFlow.js & more



04

Live Demo

Hands-on with MediaPipe

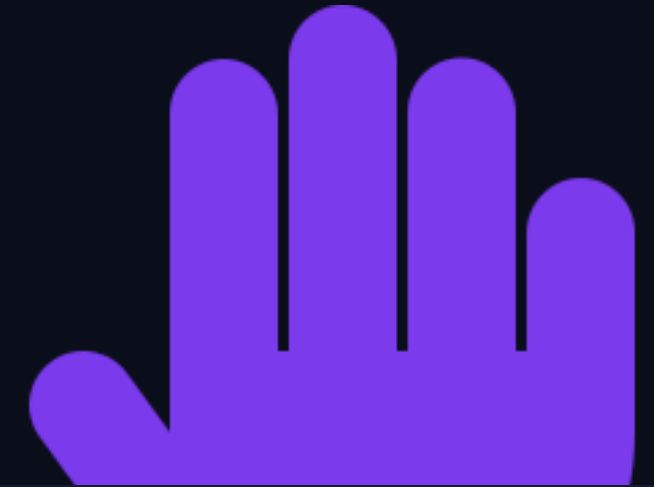


05

References

Sources cited throughout

What is Gesture-Based Hand Tracking?^[1] ^[2]



Gesture-based hand tracking is a technology that detects and follows hand movements using a camera and computer vision techniques. It allows users to interact with digital systems through natural gestures without using physical controllers or touch input.

Key Features

- **Real-time hand and finger tracking**
- **Touch-free interaction**
- **Camera-based gesture recognition**

Benefits

- **Improves accessibility and usability**
- **Creates immersive interaction experiences**
- **Reduces dependence on external devices**

How does Gesture-Based Hand Tracking work?

[1] [2]

1

Camera Input

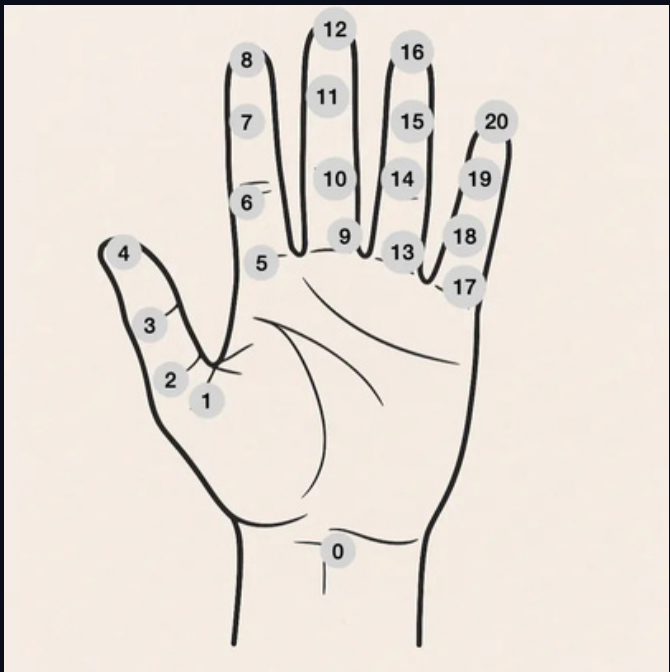
Webcam captures video frames in real time



2

AI Model

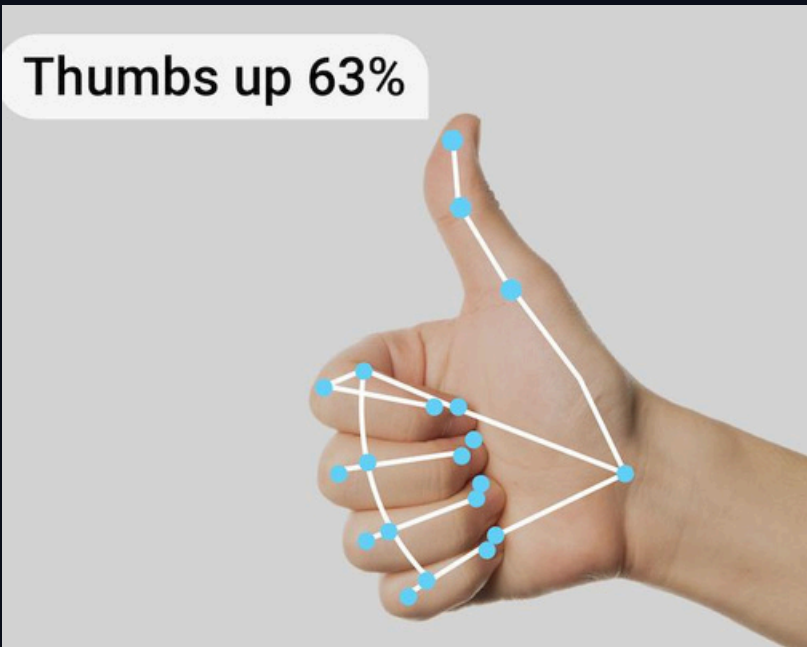
ML model detects 21 hand landmarks per hand



3

Gesture Logic

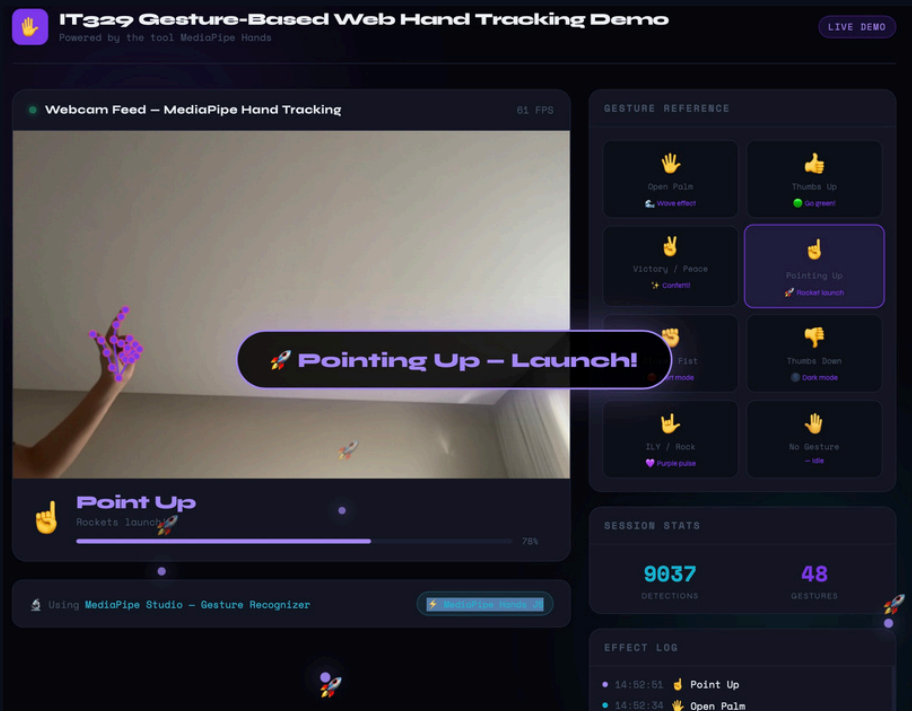
Landmark positions are mapped to gestures



4

Web Action

Browser responds to the detected gesture



How Hand Tracking Is Used on the Web



Touchless UIs

Scroll, click, and navigate websites using only hand gestures — great for accessibility and kiosks [3]



AR/VR Web Apps

Manipulate 3D objects in browser-based augmented reality without a controller [4]



Sign Language Tools

Real-time recognition of sign language gestures to assist deaf users in web communication [5]



Interactive Art

Generative web art and music apps where hand movements paint or compose in real time [3]



Health & Fitness

Web-based physiotherapy apps track hand exercises and give live feedback through the browser [4]



Gaming

Browser games that use gestures as controls — wave to jump, pinch to shoot, tilt to steer [5]

What's in the Toolbox?

MediaPipe Hands

Google

The gold standard for hand tracking on the web. Detects 21 3D landmarks per hand in real time with a lightweight ML model.

★ Best for live demos

Used for: gesture detection, hand pose

[6]

TensorFlow.js — Handpose

Google

A TensorFlow.js model that runs hand pose estimation entirely in the browser — no server needed.

★ Great for custom ML pipelines

Used for: hand landmark detection

[7]

Handtrack.js

Open Source

Simpler wrapper for detecting hands (as bounding boxes) in video streams. Easy to get started with.

★ Easiest to set up

Used for: hand presence detection

[8]

Let's Track Some Hands! 🖐️



Tool: MediaPipe Hands (JavaScript) [6]

What the demo does:

- 1 Your webcam streams video into the browser. MediaPipe reads each frame and finds your hand automatically.
- 2 MediaPipe places 21 dots on your finger joints in real time. We check which fingers are up or down to name the gesture.
- 3 Once a gesture is recognized, JavaScript changes the page — colors flash, particles explode, banners pop up.

// Snippet

```
// Every frame — is a finger up or down?
function classifyGesture(landmarks) {
  const indexUp = landmarks[8].y <
    landmarks[6].y;

  const middleUp = landmarks[12].y <
    landmarks[10].y;

  if (indexUp && middleUp) return 'Victory 🙌';
  if (indexUp && !middleUp) return 'Pointing
  👉';
}
```

Tip: smaller y = higher on screen → finger is up!



05 / References & Resources

Sources

[1]

Google AI Blog — On-Device, Real-Time Hand Tracking with MediaPipe

ai.googleblog.com/2019/08/on-device-real-time-hand-tracking.html

[2]

MediaPipe Hands Documentation

developers.google.com/mediapipe/solutions/vision/hand_landmarker

[3]

Handsfree.js — Gesture Interaction Library

handsfreejs.netlify.app

[4]

MDN — Using the WebXR Device API

developer.mozilla.org/en-US/docs/Web/API/WebXR_Device_API

[5]

TensorFlow.js — Hand Pose Detection

tensorflow.org/js/models#handpose

[6]

MediaPipe Hands JavaScript API (npm)

npmjs.com/package/@mediapipe/hands

[7]

TensorFlow.js Hand Pose Model

github.com/tensorflow/tfjs-models/tree/master/handpose

[8]

Handtrack.js — Easy Hand Detection for the Web

victordibia.com/handtrack.js/



Thank You!

